

Not Easy to Prepare a Pesto: Cryptanalysis of a Multivariate Public-Key Scheme from CCZ Equivalence

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Abstract

Multivariate cryptography is one of the challenging candidates for post-quantum cryptography. There exists a huge variety of proposals, most of them have been broken substantially. Multivariate schemes are usually constructed by applying two secret affine invertible transformations $\mathcal{S}, \mathcal{T}$ to a set of multivariate polynomials $\mathcal{F}$ (often quadratic). The secret polynomials $\mathcal{F}$ possess a trapdoor that allows the legitimate user to find a solution of the corresponding system, while the public polynomials $\mathcal{G} = \mathcal{S} \circ \mathcal{F} \circ \mathcal{T}$ look like random polynomials. In [Calderini, M., Caminata, A., Villa, I. A New Multivariate Primitive from CCZ Equivalence. J. Cryptol. 38, 25 (2025)], the authors addressed the above challenge by presenting a promising new way of constructing a multivariate scheme by considering the CCZ equivalence, which has been introduced and studied in the context of vectorial Boolean functions. The resulting proposal is called **Pesto** with security parameters s, t, m, n, q , where n is the number of variables, $s, t, m \leq n$ and q the size of the finite base field $\mathbb{F}_q$. In this paper we present an attack against **Pesto** by constructing an equivalent secret key from the public key. This attack has a precomputation phase with a complexity of

$$\max \left\{ \mathcal{O}(n^6(m-t)), \mathcal{O}\left(\frac{(m-t)(n-t)^2(n-t-s)q^s}{\mathcal{P}(q, n-t-s)}\right) \right\}$$

base field operations on average and an online complexity of

$$\mathcal{O}(q^s(m-t)(n-t-s) \cdot \min(m-t, n-t-s) + q^s t(n-t)^2 + q^s t^3)$$

base field operations to decipher a message or forge a signature, where $\mathcal{P}(q, k) := \prod_{i=1}^k (1 - 1/q^i)$. Thus, our attack breaks **Pesto** for any practical choice of the security parameters n, m, s, t, q and renders the concrete construction underlying **Pesto** insecure.

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1 Introduction

Public-key cryptography has played a vital role in securing services on the internet that we take for granted today. The security of schemes based on integer factorization and the discrete logarithm problem (DLP) is now well understood, and the related encryption algorithms have served us well over several decades. In [21] it was shown that quantum computers can solve both integer factorization and DLP in polynomial time. While large-scale quantum computers that break the actual implementations of secure internet communication are not available yet, progress is being made in constructing them. This has led the community for cryptographic research to look for new public key-primitives that are based on mathematical problems believed to be hard even for quantum computers, so called *post-quantum cryptography*. In 2016, the National Institute of Standards and Technology launched a project aimed at standardizing post-quantum public-key primitives [15]. A call for proposals was made, many candidate schemes were proposed, some winners identified and finally standardized. The candidates are based on a variety of problems, including the shortest vector problem for lattices, the problem of decoding a random linear code, or the problem of solving a system of multivariate quadratic equations over a finite field (the MQ problem).

The basic idea for multivariate schemes based on the MQ problem is to construct them around an easy to invert central mapping

$$\mathcal{F} = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n)),$$

where $f_i(x_1, \dots, x_n)$ denote multivariate polynomials in $\mathbb{F}_q[x_1, \dots, x_n]$ and $\mathbb{F}_q$ a finite field of q elements. The mapping $\mathcal{F}$ is masked with secret and randomly chosen affine mappings $\mathcal{S} : \mathbb{F}_q^m \rightarrow \mathbb{F}_q^m$ and $\mathcal{T} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$. The public key is $\mathcal{G} = \mathcal{S} \circ \mathcal{F} \circ \mathcal{T}$ and assumed to be hard to invert and thus in particular assumed to be a hard instance of the MQ problem. The first encryption scheme based on the MQ problem, named C^* , was proposed in [14] and was broken by Patarin in [18]. Since then, several multivariate encryption schemes have been proposed, for instance [5, 13, 9, 6]. The papers [7, 17, 1, 16] all present attacks on MQ-based public key encryption schemes, and as of now we are only aware of a few (e.g. [22]) that remain unbroken. Hence designing secure multivariate cryptosystems is intriguing and of high interest.

In [2] this problem is addressed and a new kind of central mapping is introduced, which can be used to construct both encryption and signature schemes. The novel feature of the central mapping g is that it is based on CCZ equivalence, which has been introduced and studied in the context of vectorial Boolean functions [4] and yields a central mapping of degree 4 and not 2, where the latter is the standard case for multivariate cryptography. Their concrete construction is called **Pesto**.

Recently, it was shown in [3] that the multivariate system of degree 4 in **Pesto** can be reduced to a quadratic system, where the reduction has complexity of $\mathcal{O}(mn^6)$ if $m \leq n$. This analysis suggested that the degree-4 system obtained via the CCZ equivalence is not more secure than a quadratic multivariate scheme. The authors of [3] did not present an attack on the reduced degree-2 system and it remains unclear whether it is efficiently solvable. Instead of attacking the MQ system, we target the decomposition problem to recover an equivalent secret key with practical complexity.

Our Contribution

We present an attack against **Pesto** with security parameters n, m, s, t, q , detailed in Section 3, by constructing an equivalent secret key, with a complexity of

$$\max \left\{ \mathcal{O}(n^6(m-t)), \mathcal{O}\left(\frac{(m-t)(n-t)^2(n-t-s)q^s}{\mathcal{P}(q, n-t-s)}\right) \right\}$$

base field operations on average, where $\mathcal{P}(q, k) := \prod_{i=1}^k (1 - 1/q^i)$, and an online complexity of

$$\mathcal{O}(q^s(m-t)(n-t-s) \cdot \min(m-t, n-t-s) + q^s t(n-t)^2 + q^s t^3)$$

base field operations. Thereby $t \leq \min(n, m)$ and $1 \leq s \leq n-t$.

Note that we have the approximation

$$\mathcal{P}(q, k) \approx \exp\left(\frac{1}{1-q}\right)$$

by writing $\ln(\mathcal{P}(q, k)) = \sum_{i=1}^k \ln(1 - 1/q^i) \approx \sum_{i=1}^k -(1/q)^i$ and using the limit of the geometric series as well as the approximation $\ln(1 - (\frac{1}{q})^i) \approx -(\frac{1}{q})^i$ since $(\frac{1}{q})^i$ is small enough. Hence, the term $\mathcal{P}(q, n-t-s)$ is close to 1 for the instances of **Pesto** we are investigating.

Typical sizes for the security parameters proposed by the designers are $n = 27, m = 25, t = 10, s = 2, \mathbb{F}_q = \mathbb{F}_{2^6}$, and thus this attack especially renders practical constructions underlying **Pesto** insecure.

Organization

In Section 2, we introduce the notation used, recall some facts from the theory of quadratic forms and develop the theory for our attack. In Section 3, we give a description of **Pesto**, and we outline our attack in Section 4. We conclude the paper by discussing the lessons learned and giving some directions to conduct further research to build practical and possibly secure systems based on this novel approach.

2 Preliminaries

We assume that the reader is familiar with basic facts on finite fields. For more details, [12] is a suitable reference. Throughout this work, let n and m be positive integers. The finite field with q elements, q a prime power is denoted by $\mathbb{F}_q$. With $\mathbb{F}_q^n$ we denote the vector space of dimension n over $\mathbb{F}_q$ and with $|\cdot|$ the cardinality of a set. If not stated otherwise an element of $\mathbb{F}_q^n$ is a column vector. The canonical unit basis of $\mathbb{F}_q^n$ is denoted with $e_1 = (1, 0, \dots, 0)^T, \dots, e_n = (0, \dots, 0, 1)^T$, T the transpose. The subspace generated by, not necessarily linear independent, elements $v_1, \dots, v_l \in \mathbb{F}_q^n$ is denoted by $\langle v_1, \dots, v_l \rangle$. The multivariate polynomial ring in n unknowns over $\mathbb{F}_q$ is denoted by $\mathbb{F}_q[x_1, \dots, x_n]$. By $M(n \times m, \mathbb{F}_q)$, we denote the set of $n \times m$ matrices and by Id the identity matrix in case of $n = m$. For two sets X, Y , the relation

$X \subset Y$ also includes equality. With $a \ll b$ for $a, b \in \mathbb{R}$ we denote that a is much smaller than b , where it will be always clear from that context what being "much smaller" means.

Let F be a function $F: \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$. Notice that F can be represented as $F = (f_1, \dots, f_m)$ with $f_i: \mathbb{F}_q^n \rightarrow \mathbb{F}_q, 1 \leq i \leq m$. We call f_i the i -th coordinate of F and denote it also by F_i . To represent F we can extend the so called algebraic normal form (ANF) as follows: Any function $F: \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ has a unique representation of the form $F(x) = \sum_{u \in \mathbb{N}^n} a_u x_1^{u_1} \cdots x_n^{u_n} \bmod x_1^q - x_1, \dots, x_n^q - x_n$ with $a_u \in \mathbb{F}_q$. The unique representation of F with all $u_1, \dots, u_n$ less than q is called the *algebraic normal form (ANF)*. This can be extended to an ANF for $F: \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ by setting $F(x) = \sum_{u \in \mathbb{N}^n} a_u x_1^{u_1} \cdots x_n^{u_n}$ with $a_u \in \mathbb{F}_q^m$ and all $u_1, \dots, u_n$ less than q . The monomial $x_1^{u_1} \cdots x_n^{u_n}$ belongs to a *term* of F if and only if $a_u \neq 0$. The *degree* of F is defined as $\deg(F) = \max\{\sum_{i=1}^n u_i \mid u \in \{0, \dots, q-1\}^n, a_u \neq 0\}$. If $F = 0$, we define $\deg(F) = -\infty$. We call F *affine* if $\deg(F) \leq 1$, *linear* if additionally $F(0) = 0$, and *quadratic* if $\deg(F) = 2$. If all terms have the same degree then F is called *homogeneous*. If a function S is linear or affine it can be written as $S(x) = Ax$ or $S(x) = Ax + d$ respectively, where $A \in M(n \times m, \mathbb{F}_q)$ and $d \in \mathbb{F}_q^m$.

We will now give some well-known facts about quadratic forms necessary for our attacks. Proofs will be only given if these enhance the understanding of the attack. For more details we refer to [11] and [12].

Definition 2.1. A function $q: \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ is called a quadratic form if

1. $q(ax) = a^2 q(x)$ holds for any scalar $a \in \mathbb{F}_q$, and
2. the function

$$\begin{aligned} \mathfrak{B}_q: \mathbb{F}_q^n \times \mathbb{F}_q^n &\rightarrow \mathbb{F}_q \\ (x, y) &\mapsto q(x + y) - q(x) - q(y) \end{aligned}$$

is a symmetric bilinear form.

Remark 2.1. The ANF of a quadratic form $q \neq 0$ is a homogeneous quadratic polynomial. In case of $\mathbb{F}_q = \mathbb{F}_2$ we have, by abuse of notation, to set $x_i = x_i^2$ in q for the above statement to become true.

Definition 2.2. Given a quadratic form $q: \mathbb{F}_q^n \rightarrow \mathbb{F}_q$.

1. Two vectors $x, y \in \mathbb{F}_q^n$ are called orthogonal if $\mathfrak{B}_q(x, y) = 0$.
2. For $X \subset \mathbb{F}_q^n$, we set $X^\perp := \{y \in \mathbb{F}_q^n \mid \mathfrak{B}_q(x, y) = 0 \text{ for all } x \in X\}$. $X^\perp$ is called the orthogonal complement of X in $\mathbb{F}_q^n$.
3. If q is even we always have $\mathfrak{B}_q(x, x) = 0$. Such a form is called symplectic.
4. The radical of $\mathfrak{B}_q$ is defined as $\text{rad}(\mathfrak{B}_q) := \mathbb{F}_q^{n\perp}$.
5. The rank of $\mathfrak{B}_q$ is defined as $\text{rank}(\mathfrak{B}_q) := n - \dim(\text{rad}(\mathfrak{B}_q))$. The rank of q is defined as $\text{rank}(q) := \text{rank}(\mathfrak{B}_q)$.
6. q is called isotropic if there exists a non-zero vector $v \in \mathbb{F}_q^n$ with $q(v) = 0$. Such v is called an isotropic vector.

Remark 2.2. It is $q(x + y) = q(x) + q(y)$ for all $x \in \mathbb{F}_q^n$ and $y \in \text{rad}(\mathfrak{B}_q)$. In particular, q is a linear form when restricted to $\text{rad}(\mathfrak{B}_q)$. The radical will play a crucial role in our attack.

Two quadratic forms $\mathbf{q}, \mathbf{q}'$ over $\mathbb{F}_q^n$ are called *isometric* if there exists a bijective linear mapping L with $\mathbf{q}(x) = \mathbf{q}'(L(x))$. In the same vein isometry between symmetric bilinear forms is defined. Isometry yields an equivalence relation for forms and its associated bilinear forms. We have the following obvious lemma.

Lemma 2.3. *Given two isometric forms $\mathbf{q}$ and $\mathbf{q}'$ with $\mathbf{q} = \mathbf{q}'(L(x))$, then $\text{rad}(\mathfrak{B}_{\mathbf{q}'}) = L(\text{rad}(\mathfrak{B}_{\mathbf{q}}))$.*

The following corollary will be crucial for our attack.

Corollary 2.4. *Given quadratic forms $\mathbf{q}'_1, \dots, \mathbf{q}'_l$ over $\mathbb{F}_q^n$, where each form has at most r indeterminates $x_1, \dots, x_r$. For a bijective linear mapping L , consider the isometric forms $\mathbf{q}_1(x) = \mathbf{q}'_1(L(x)), \dots, \mathbf{q}_l(x) = \mathbf{q}'_l(L(x))$. Let $\alpha_1, \dots, \alpha_l \in \mathbb{F}_q$. Then,*

1. $\text{rad}(\mathfrak{B}_{\mathbf{q}_i}) \supset L^{-1}(\langle e_{r+1}, \dots, e_n \rangle)$ for $i = 1, \dots, l$.
2. $\bigcap_{i=1}^l \text{rad}(\mathfrak{B}_{\mathbf{q}_i}) \subset \text{rad}(\mathfrak{B}_{\alpha_1 \mathbf{q}_1 + \dots + \alpha_l \mathbf{q}_l})$.
3. $L^{-1}(\langle e_{r+1}, \dots, e_n \rangle) \subset \text{rad}(\mathfrak{B}_{\alpha_1 \mathbf{q}_1 + \dots + \alpha_l \mathbf{q}_l})$.
4. $\text{rank}(\mathfrak{B}_{\alpha_1 \mathbf{q}_1 + \dots + \alpha_l \mathbf{q}_l}) \leq r$.

Proof. The first statement is a direct consequence of Lemma 2.3 and the second statement follows from the definition of the radical. The last two statements follow easily from the first two. $\square$

If q is odd then any quadratic form $\mathbf{q}(x) = \sum_{i,j} \alpha_{ij} x_i x_j$ can be uniquely written as $x^T A x$, $A \in M(n \times n, \mathbb{F}_q)$, with $a_{ij} = a_{ji} = \frac{\alpha_{ij} + \alpha_{ji}}{2}$, $i \neq j$, and $a_{ii} = \alpha_{ii}$. It is $\mathfrak{B}_{\mathbf{q}}(x, y) = 2 \cdot x^T A y$ and thus $\text{rank}(\mathfrak{B}_{\mathbf{q}}) = \text{rank}(A)$.

The situation is more subtle for q even. It is still possible to represent $\mathbf{q}(x) = \sum_{i,j} \alpha_{ij} x_i x_j$ as $x^T A x$, but A is no longer symmetric and unique. For instance A could be chosen to be an upper triangular matrix U starting with $\alpha_{1,1}, \alpha_{1,2}, \dots$ in the first row, $\alpha_{2,2}, \dots$ in the second row and so forth but also as the lower triangular matrix $A = U^T$. In any case $\mathfrak{B}_{\mathbf{q}}(x, y) = x^T (A + A^T) y$, where now $(A + A^T)$ is uniquely determined, i.e., independent from the choice of A and skew-symmetric. Again $\text{rank}(\mathfrak{B}_{\mathbf{q}}) = \text{rank}(A + A^T)$. As a consequence we have:

Theorem 2.5. *Given $\mathbb{F}_q^n$ for a fixed n . For a uniformly at random chosen quadratic form $\mathbf{q}$ over $\mathbb{F}_q^n$ we have $\text{rank}(\mathfrak{B}_{\mathbf{q}}) = r$, where r is distributed as the rank of a uniformly at random chosen symmetric matrix in $M(n \times n, \mathbb{F}_q)$, with zero diagonal in case of q even.*

Remark 2.3. *Given $\mathbb{F}_q^n$ for a fixed n . For a uniformly at random chosen quadratic form $\mathbf{q}$ over $\mathbb{F}_q^n$, we expect that its rank is equal to n with high probability.*

This statement is justified as follows: Randomly chosen symmetric matrices with zero diagonal in case of q even behave randomly enough in the set $M(n \times n, \mathbb{F}_q)$ and thus its rank is similarly distributed. Therefore, by applying the theorem statement, the claim follows. More precisely, we expect the probability to be the same as the probability of an element in $M(n \times n, \mathbb{F}_q)$ to be invertible,

which is equal to $\mathcal{P}(q, n) := \prod_{i=1}^n (1 - 1/q^i)$, see, e.g., Chapter 7 in [8]. The above theorem and remark will be of great importance for our attack.

The next lemma and the subsequent theorem, which is related to the so-called *OV polynomials* explained in more detail in Section 3, will be of importance in Subsection 4.3.

Lemma 2.6. *Let $X = \{x_1, \dots, x_v\}$, $Y = \{x_{v+1}, \dots, x_n\}$ and L a bijective linear mapping on $\mathbb{F}_q^n$. Given uniformly at random chosen quadratic forms $\mathbf{q}_\nu: \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ of the shape*

$$\sum_{1 \leq i \leq j \leq v} \alpha_{ij}^{(\nu)} x_i x_j + \sum_{\substack{1 \leq i \leq v \\ v+1 \leq j \leq n}} \beta_{ij}^{(\nu)} x_i x_j = \sum_{1 \leq i \leq j \leq v} \alpha_{ij}^{(\nu)} x_i x_j + \sum_{1 \leq i \leq v} x_i \ell_i^{(\nu)}(x_{v+1}, \dots, x_n), \quad (1)$$

where $1 \leq \nu \leq m$ and the corresponding isometric forms $\mathbf{q}'_\nu(x) := \mathbf{q}_\nu(L(x))$. Then we have:

1. $I_0 := \underbrace{\{0\} \times \dots \times \{0\}}_{\times v} \times \underbrace{\mathbb{F}_q \times \dots \times \mathbb{F}_q}_{\times n-v} \subset I_{\mathbf{q}_\nu}$ for $1 \leq \nu \leq m$, where $I_{\mathbf{q}_\nu}$ denotes the union of $\{0\}$ and the set of isotropic vectors of $\mathbf{q}_\nu$.
2. If $n - v \leq m$, then for $u \in \mathbb{F}_q^v \setminus \{0\}$ the number of isotropic vectors $(u, w) \in \cap_{\nu=1}^m I_{\mathbf{q}_\nu}$ is less than one on average.
3. Let $\mathcal{B} = \{b_1, \dots, b_{n-v}\}$ be a linear independent subset contained in $L^{-1}(I_0)$ and $u_1, \dots, u_v$ an extension of $\mathcal{B}$ to a basis of $\mathbb{F}_q^n$. Then for the linear mapping L' defined by the matrix $(u_1, \dots, u_v, b_1, \dots, b_{n-v})$, the composition $L \circ L'$ has the representation matrix

$$\begin{pmatrix} A & 0 \\ B & C \end{pmatrix}$$

with respect to the canonical basis. Thereby $A \in M(v \times v, \mathbb{F}_q)$, $B \in M((n-v) \times v, \mathbb{F}_q)$, $C \in M((n-v) \times (n-v), \mathbb{F}_q)$.

4. The forms $\mathbf{q}'_\nu(L'(x))$ contain no quadratic monomials of the form $x_i x_j$ with $x_i, x_j \in Y$.

Proof. The first statement is obvious due to the special shape of each $\mathbf{q}_\nu$. The second statement can be seen as follows: Fixing the first v variables conforming to u in each $\mathbf{q}_\nu$ yields m linear equations in the remaining $n - v$ variables, which readily becomes apparent from Expression 1. As $u \neq 0$, it is expected that the resulting system of equations has rank $n - v$ if $n - v \leq m$ and thus at most one solution. Hence the amount of isotropic vectors $(u, w) \in \cap_{\nu=1}^m I_{\mathbf{q}_\nu}$ for fixed non-zero u is at most one on average.

Since L maps $\mathcal{B}$ into I_0 the third statement follows immediately. The fourth statement follows from the shape of the representation matrix of $L \circ L'$. $\square$

Remark 2.4. *Note that we expect $|\cap_{\nu=1}^m (I_{\mathbf{q}_\nu} \setminus I_0)| \leq q^v - 1$ by 1. and 2..*

Theorem 2.7. *Given $\mathbf{q}_\nu(L(x))$, $1 \leq \nu \leq m$ as in Lemma 2.6 with $n - v \leq m$. If $\left(\frac{q^v - 1}{q^n}\right) \left(\frac{(n-v)q^v}{\mathcal{P}(q, n-v)}\right) \ll 1$ then L' as in Lemma 2.6 can be computed with $\mathcal{O}\left(\frac{m(n-v)q^v}{\mathcal{P}(q, n-v)}\right)$ quadratic form evaluations.*

Proof. To compute L' one chooses b uniformly at random from $\mathbb{F}_q^n$ and checks whether $\mathbf{q}_\nu(L(b)) = 0$ for all $1 \leq \nu \leq m$. Repeating this approximately $(n - v)q^v \cdot \mathcal{P}(q, n - v)^{-1}$ times yields $n - v$ linearly independent isotropic vectors $b_1, \dots, b_{n-v}$ contained in $L^{-1}(I_0)$ on average, as long as

$$((q^v - 1)/q^n) ((n - v)q^v \mathcal{P}(q, n - v)^{-1}) \ll 1.$$

This can be seen as follows. The probability that $\{b_1, \dots, b_{n-v}\} \subset L^{-1}(I_0)$ is linear independent is equivalent to randomly choosing a basis in $\mathbb{F}_q^{n-v}$ due to the shape of I_0 . The latter is equivalent to randomly choosing an invertible $(n - v) \times (n - v)$ -matrix over $\mathbb{F}_q$; the probability for this is $\mathcal{P}(q, n - v) = (1 - \frac{1}{q})(1 - \frac{1}{q^2}) \cdots (1 - \frac{1}{q^{n-v}})$. It could have happened that some b was chosen from $L^{-1}(I_{\mathbf{q}_\nu} \setminus I_0)$ for $1 \leq \nu \leq n - v$. Such an isotropic vector is chosen with probability at most $\frac{q^v - 1}{q^n}$ by Remark 2.4. Since by assumption

$$((q^v - 1)/q^n) ((n - v)q^v \mathcal{P}(q, n - v)^{-1}) \ll 1,$$

this is not expected to happen at all on average.

Thus, after $(n - v)q^v \mathcal{P}(q, n - v)^{-1}$ trials, $\mathcal{B}$ and finally L' is found on average. Since the check if $b \in L^{-1}(I_0)$ is carried out by checking that b is isotropic for all m forms, we get the desired number of evaluations and the theorem follows. Note that the cost for checking the linear independence of the found isotropic vectors is dominated by the quadratic form evaluations; each of them being in $\mathcal{O}(n^2)$. $\square$

For estimating the complexity of our attack, similarly as in [2], we use that solving a linear system of m equations in n unknowns can be performed in $\mathcal{O}(mn \min(m, n))$ base field operations.

3 A Description of Pesto

In this section we describe the design of **Pesto**. The approach makes use of CCZ equivalence. As this fact is neither of importance for our attack nor for understanding **Pesto** we refer for the details to [2]. We mainly follow the notation from the designers' work.

Definition 3.1. *Given s, t, n . For a polynomial in n indeterminates, we denote them with $x_1, \dots, x_t, y_1, \dots, y_{n-t}$. An Oil and Vinegar (OV) polynomial is a polynomial of the form*

$$\sum_{j,k \in V} \alpha_{jk} z_j z_k + \sum_{j \in V, k \in O} \beta_{jk} z_j z_k + \sum_{j=1}^n \gamma_j z_j + \delta \quad (2)$$

with $\{z_j \mid j \in V\} = \{x_1, \dots, x_t, y_1, \dots, y_s\}$, $\{z_j \mid j \in O\} = \{y_{s+1}, \dots, y_{n-t}\}$ and $\alpha_{jk}, \beta_{jk}, \gamma_j, \delta \in \mathbb{F}_q$. The set $\{z_j \mid j \in V\}$ is called the set of vinegar variables and the set $\{z_j \mid j \in O\}$ the set of oil variables. If $|V| > |O|$, the OV polynomial is also referred to as an unbalanced OV polynomial.

Remark 3.1. *When fixing the vinegar variables, an OV polynomial is linear in the oil variables. Therefore, if $|V|$ is small enough, we can find the zeros efficiently by first guessing the vinegar variables and then applying Gaussian elimination.*

For more details on OV polynomials and early approaches to build cryptosystems with them, see e.g., [20],[9],[10].

We now fix positive integers n, m, t, s with $t \leq \min(n, m)$ and $s \leq n - t$. The central map is based on the following maps:

- For a randomly chosen quadratic map $\mathbf{q}: \mathbb{F}_q^{n-t} \rightarrow \mathbb{F}_q^t$ set $x - \mathbf{q}(y) := (x_1 - \mathbf{q}_1(y), \dots, x_t - \mathbf{q}_t(y))$. Note that using $-$ instead of $+$ is due to the notation in [2].
- $U: \mathbb{F}_q^n \rightarrow \mathbb{F}_q^{m-t}$ is a mapping in which the $m - t$ coordinates consists of randomly chosen OV polynomials conforming Definition 3.1.

Let

- $A_1: \mathbb{F}_q^m \rightarrow \mathbb{F}_q^m, x \mapsto S(x) + c, S \in M(m \times m, \mathbb{F}_q), c \in \mathbb{F}_q^m$ be a randomly chosen affine bijection.
- $A_2: \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n, x \mapsto T(x) + d, T \in M(n \times n, \mathbb{F}_q), d \in \mathbb{F}_q^n$ be a randomly chosen affine bijection.

Set $G := (x - \mathbf{q}(y), U(x - \mathbf{q}(y), y))$. The map

$$G_{\text{pub}} := A_1 \circ G \circ A_2$$

is the public key, and the tuple $(A_1, A_2, \mathbf{q}, U)$ is the secret key.

Pesto Encryption Scheme

Encryption.

1. The sender would like to encrypt the message $\mathbf{m}$. To do so some redundancy r has to be added, such that the actual message to be encrypted is $z = (\mathbf{m}||r) \in \mathbb{F}_q^n$ with $||$ denoting the concatenation. The necessity for the redundancy r will become clear during the decryption process. For more details on how to choose the redundancy see, e.g., [19].
2. The sender encrypts z by computing $c = G_{\text{pub}}(z) \in \mathbb{F}_q^m$.

Decryption.

1. From $c \in \mathbb{F}_q^m$ compute $c' = A_1^{-1}(c)$.
2. For simplicity set $z' = A_2(z)$ where z is the sought for plaintext. So $c' = G(z')$. c' can be decomposed into $c' = (c'_Q, c'_U) \in \mathbb{F}_q^t \times \mathbb{F}_q^{m-t}$. Similarly $z' = (x', y') \in \mathbb{F}_q^t \times \mathbb{F}_q^{n-t}$. It holds

$$\begin{aligned} c'_Q &= x' - \mathbf{q}(y') \\ c'_U &= U(x' - \mathbf{q}(y'), y') = U(c'_Q, y'). \end{aligned}$$

For fixed values $y'_1, \dots, y'_s$ the system of equations $c'_U = U(c'_Q, y')$ becomes linear in $y'_{s+1}, \dots, y'_{n-t}$ as the coordinates of U are OV polynomials. By exhaustive search over (possibly) all $(y'_1, \dots, y'_s) \in \mathbb{F}_q^s$, a set of values $\mathcal{Y}' \subset \mathbb{F}_q^{n-t}$ is computed by solving the resulting linear system of equations.

3. For all $y \in \mathcal{Y}'$, the corresponding value x' is easily computed from $x' - \mathbf{q}(y')$ and finally all possible plaintexts $z = A_2^{-1}(z') = A_2^{-1}((x', y'))$. The correct plaintext is identified by the redundancy r .

Pesto Signature Scheme

This scheme can easily be turned into a signature scheme. The hash value $H(\mathbf{m})$ of the message $\mathbf{m}$ to be signed gets the role of c . From this all solutions z with $H(\mathbf{m}) = G_{\text{pub}}(z)$ are computed and one is chosen. Checking the signature is done by verifying that $H(\mathbf{m}) = G_{\text{pub}}(z)$.

To ensure that $H(\mathbf{m}) = G_{\text{pub}}(z)$ has a solution, one approach is for instance to consider $\mathbf{m} = (\mathbf{m}' || r)$, where r is a random value. By this variation, the system is expected to become solvable at some point.

Parameter Choices

Concrete parameters are, for instance, $n = 27, m = 25, t = 10, s = 2$, and $q = 2^6$ for the signature scheme. The authors' security analysis advises that in general s has to be chosen such that $s > 0$ and $t \approx \frac{n}{3}$. For more details on that parameter choices see [2].

There also exists a kind of compressed version of **Pesto**, where the linear part of A_1 is of the form

$$\begin{pmatrix} A & 0 \\ B & C \end{pmatrix}$$

$A \in M(t \times t, \mathbb{F}_q), B \in M((n-t) \times t, \mathbb{F}_q), C \in M((n-t) \times (n-t), \mathbb{F}_q)$. This leads to a reduction in the key size by not changing the security (see Remark 5 in [2]). The latter fact will also be implicitly verified by our attack.

4 The Attack

The attack exploits the fact that it is possible to transform G_{pub} into a set of polynomials, which is equally easy to invert as the central map. To simplify the notation, let (x, y) denote the vector of indeterminates $(x_1, \dots, x_t, y_1, \dots, y_{n-t})$ or an element of $\mathbb{F}_q^n$. The meaning will always be clear from the context. Let $q'_i(x, y)$ denote the i -th coordinate of $(x - \mathbf{q}(y)) \circ A_2(x, y)$ and

$$Q' := \langle q'_1(x, y), \dots, q'_t(x, y) \rangle$$

the subspace of $\mathbb{F}_q[x_1, \dots, x_n]$ considered as a vector space.

4.1 Step 1: Recover the Space Q'

Let $G' := G \circ A_2$ and g'_i its coordinates. Similarly, let $g_{\text{pub},i}$ denote the coordinates of the public key. With this notation, $Q' = \langle g'_1, \dots, g'_t \rangle$. Step 1 can be omitted for the compressed version as in this case, we have $Q' = \langle g_{\text{pub},1}, \dots, g_{\text{pub},t} \rangle$ already. How to recover Q' is also given in [2] but to keep the paper self contained, we will show it here as well and additionally give more details for why it works. To recover Q' we compute all linear combinations

$$\sum_{i=1}^m \alpha_i g_{\text{pub},i} = \sum_{i=1}^m \alpha_i (A_1 \circ G')_i$$

such that the sum has degree at most 2. We claim the following.

Claim 4.1. *The polynomials $g'_{t+1}, \dots, g'_m$ can be considered as randomly chosen polynomials of degree 4. Hence any linear combination has still degree 4 almost for sure.*

The second part of this claim is justified by the following argument. There are

$$\binom{4+n-1}{n-1} + \binom{3+n-1}{n-1} = \binom{4+n-1}{4} + \binom{3+n-1}{3} = \mathcal{O}(n^4)$$

monomials of degree 3 and 4 in n variables. It is

$$\sum_{i=1}^m \alpha_i (A_1 \circ G')_i = \sum_{i=1}^m \alpha'_i g'_i,$$

for properly chosen α'_i . Hence the sum can only have a degree of at most 2 if the terms of degree 3 and 4 in the partial sum $\sum_{t+1}^m \alpha'_i g'_i$ cancel out. Since these are assumed to behave like randomly chosen and m is much lower than n^4 for values m, n that lead to practical instances of **Pesto**, the parts of degree 3 and 4 of $g'_{t+1}, \dots, g'_m$ will be linear independent almost for sure. From this we get that $\alpha = (\alpha_1, \dots, \alpha_m)$ is a solution if and only if $\alpha S \in \mathbb{F}_q^t \times \{0\} \times \dots \times \{0\} \subset \mathbb{F}_q^m$. Thus the solution space is generated by the first t rows of S^{-1} and will therefore be denoted by $S_{Q'}^{-1}$. We denote a basis of $S_{Q'}^{-1}$ by $\alpha_{Q',1}, \dots, \alpha_{Q',t} \in \mathbb{F}_q^m$. Such a basis can be computed by solving a system of linear equations with m unknowns and at most $\binom{4+n-1}{n-1} + \binom{3+n-1}{n-1}$ equations, i.e. one equation for each monomial of degree 4 and 3 occurring in at least one of the $g_{\text{pub},i}$.

Let $S^{-1'}$ denote the matrix, where the first rows are $\alpha_{Q',1}, \dots, \alpha_{Q',t}$ and the last $m-t$ rows are the vectors $e_{t+1}, \dots, e_m$. Then we denote first t components of $S^{-1'} \circ G_{\text{pub}}$ by $q'_{\alpha_{Q',1}}, \dots, q'_{\alpha_{Q',t}}$, and we have

$$Q' = \langle q'_{\alpha_{Q',1}}, \dots, q'_{\alpha_{Q',t}} \rangle.$$

For the compressed version, we can simply take $S^{-1'} = Id$.

This step is dominated by the complexity of computing $\alpha_{Q',1}, \dots, \alpha_{Q',t}$ by Gaussian elimination and has therefore a worst-case complexity of

$$\mathcal{O}\left(m^2 \left(\binom{4+n-1}{n-1} + \binom{3+n-1}{n-1} \right)\right) = \mathcal{O}(m^2 n^4).$$

4.2 Step 2: Transform $q'_{\alpha_{Q',i}}$ Into $\mathfrak{l}_i(x) - \mathfrak{q}_i''(y)$

Consider the homogeneous part of each $q'_{\alpha_{Q',i}}$ of degree 2, denoted $q'_{h,\alpha_{Q',i}}$. These are linear combinations of the homogeneous parts of $q'_i(x, y)$, which are equal to the homogeneous parts of degree 2 of $\mathfrak{q}_i(T(x, y))$, $i = 1, \dots, t$. So, each $q'_{h,\alpha_{Q',i}}$ is isometric to a linear combination of $\mathfrak{q}_{h,1}, \dots, \mathfrak{q}_{h,t}$, where $\mathfrak{q}_{h,i}$ denotes the homogeneous part of $\mathfrak{q}_i$. Note that we have a slight abuse of notation here, as by definition the $\mathfrak{q}_i$ are mappings from $\mathbb{F}_q^{n-t}$ to $\mathbb{F}_q$ and we now interpret

them as mappings from $\mathbb{F}_q^n$ to $\mathbb{F}_q$ by setting the first indeterminates to zero. In particular, $\langle e_1, \dots, e_t \rangle \subset \text{rad}(\mathfrak{B}_{\mathfrak{q}_{h,i}})$. By Remark 2.3 we expect that

$$\cap_{i=1}^t \text{rad}(\mathfrak{B}_{\mathfrak{q}_{i,h}}) = \langle e_1, \dots, e_t \rangle$$

almost for sure (as t is large enough).

At first the radical of $\mathfrak{B}_{q'_{h,\alpha_{Q'},1}}$ is computed. Since $q'_{h,\alpha_{Q'},1}$ is isometric to a linear combination of $\mathfrak{q}_{h,1}, \dots, \mathfrak{q}_{h,t}$ we expect by Corollary 2.4 and what we have discussed so far that $\text{rad}(\mathfrak{B}_{h,q'_{\alpha_{Q'},1}}) = \langle T^{-1}(e_1), \dots, T^{-1}(e_t) \rangle$ and thus to have dimension t . If the dimension is higher, it will be intersected with $\text{rad}(\mathfrak{B}_{q'_{h,\alpha_{Q'},2}})$, and so on. We expect due to the above mentioned isometry that at latest

$$\cap_{i=1}^t \text{rad}(\mathfrak{B}_{q'_{h,\alpha_{Q'},i}}) = \langle T^{-1}(e_1), \dots, T^{-1}(e_t) \rangle$$

almost for sure.

To compute the radical of $\mathfrak{B}_{q'_{h,\alpha_{Q'},i}}$ one has to solve the system of linear equations

$$\begin{aligned} \mathfrak{B}_{q'_{h,\alpha_{Q'},i}}(e_1, y) &= 0 \\ &\vdots \\ \mathfrak{B}_{q'_{h,\alpha_{Q'},i}}(e_n, y) &= 0. \end{aligned}$$

Hence in the worst-case, i.e., one has to compute the complete intersection, a system of tn equations in n unknowns has to be solved. We conclude that this part has complexity $\mathcal{O}(tn^3)$.

Once a basis $t'_1, \dots, t'_t$ of $\langle T^{-1}(e_1), \dots, T^{-1}(e_t) \rangle$ is recovered we generate a matrix $T^{-1'}$, where the t first columns are $t'_1, \dots, t'_t$ and the last $n-t$ columns $t'_{t+1}, \dots, t'_n$ are chosen such that $t'_1, \dots, t'_n$ form as basis of $\mathbb{F}_q^n$. Then the composition $TT^{-1'}$ has the shape

$$\begin{pmatrix} A & B \\ 0 & C \end{pmatrix}.$$

Thereby $A \in M(t \times t, \mathbb{F}_q), B \in M(t \times (n-t), \mathbb{F}_q), C \in M((n-t) \times (n-t), \mathbb{F}_q)$. The composition $g''_i := q'_{\alpha_{Q'},i} \circ T^{-1'}$ has the shape $\mathfrak{l}_i(x) - \mathfrak{q}''_i(y)$, where $\mathfrak{q}''_i(y)$ is a quadratic mapping from $\mathbb{F}_q^{n-t} \rightarrow \mathbb{F}_q$ to $\mathbb{F}_q$ and $\mathfrak{l}_i$ a linear mapping from $\mathbb{F}_q^t$ to $\mathbb{F}_q$. Note that A is necessarily invertible and thus given $y = (y_1, \dots, y_{n-t})$ and $c'_1, \dots, c'_t$ there always exists $x = (x_1, \dots, x_t)$ such that $g''_1(x, y) = c'_1, \dots, g''_t(x, y) = c'_t$. In other words $(g''_1(\cdot, y), \dots, g''_t(\cdot, y))$ is still a bijection over $\mathbb{F}_q^t$ for any $y \in \mathbb{F}_q^{n-t}$. This fact is of importance later.

Then $S^{-1'} \circ G_{\text{pub}} \circ T^{-1'}$ is equal to

$$G'' := \begin{pmatrix} g''_1 \\ \vdots \\ g''_t \\ g''_{t+1} \\ \vdots \\ g''_m \end{pmatrix} = \begin{pmatrix} g''_1 \\ \vdots \\ g''_t \\ g_{\text{pub},t+1} \circ T^{-1'} \\ \vdots \\ g_{\text{pub},m} \circ T^{-1'} \end{pmatrix}.$$

The current step is dominated by computing the radical(s) and thus has a complexity of $\mathcal{O}(tn^3)$.

Remark 4.1. This step already reduces the entropy of the system by t . Indeed, intercepted a ciphertext c , denote $c' := S^{-1'}(c)$. For the first t entries one gets the equations $c'_i = g''_i = \mathbf{l}_i(x) - \mathbf{q}''_i(y)$. Exhaustive search over the $n - t$ variables y yields a linear system of equations in the remaining t variables x . Solving it and checking the result against the whole ciphertext yields the message after applying $T^{-1'}$ to the left-over solution.

In the next step we will transform $g_{\text{pub},t+1} \circ T^{-1'}, \dots, g_{\text{pub},m} \circ T^{-1'}$ into polynomials such that, combined with g''_i , decryption will be as easy as with the knowledge of A_1, A_2 .

4.3 Step 3: Transform the Lower Part of G''

Since the $q'_{\alpha_{Q',i}}$ generate Q' and thus especially the elements $q'_i(x, y)$ we have, for properly chosen elements $\alpha_{jkl}, \beta_{jkl}, \gamma_{ijl}, \delta_{il}, \nu_{jl}, \mu_l$, the identity

$$\begin{aligned} & \left(U(x - \mathbf{q}(y), y) \circ A_2 \circ T^{-1'} \right)_l = \\ & \sum_{1 \leq j \leq k \leq t} \alpha_{jkl} g''_j g''_k + \\ & \sum_{j \in \{1, \dots, t\}, k \in \{1, \dots, n-t\}} \beta_{jkl} g''_j y_k + \\ & \sum_{1 \leq i \leq j \leq n-t} \gamma_{ijl} y_i y_j + \\ & \sum_{i=1}^t \delta_{il} g''_i + \\ & \sum_{j=1}^{n-t} \nu_{jl} y_j + \\ & \mu_l, \end{aligned}$$

where l denotes the l -th coordinate for $l = 1, \dots, t$. As $g''_{t+1}, \dots, g''_m$ are linear combinations of $\left(U(x - \mathbf{q}(y), y) \circ A_2 \circ T^{-1'} \right)_l$ and $g''_1, \dots, g''_t$ due to the composition with A_1 , we get from the above identity that

$$\begin{aligned} g''_l = & \sum_{1 \leq j \leq k \leq t} \alpha'_{jkl} g''_j g''_k + \\ & \sum_{j \in \{1, \dots, t\}, k \in \{1, \dots, n-t\}} \beta'_{jkl} g''_j y_k + \\ & \sum_{1 \leq i \leq j \leq n-t} \gamma'_{ijl} y_i y_j + \\ & \sum_{i=1}^t \delta'_{il} g''_i + \\ & \sum_{j=1}^{n-t} \nu'_{jl} y_j + \\ & \mu'_l, \end{aligned} \tag{3}$$

for $l = t + 1, \dots, m$ and properly chosen $\alpha'_{jkl}, \beta'_{jkl}, \gamma'_{ijl}, \delta'_{il}, \nu'_{jl}, \mu'_l$.

This solution is unique if we exclude exceptional cases over $\mathbb{F}_2$, which are not of interest for this paper anyway. This can be seen as follows: Consider the difference for two different solutions, i.e.,

$$\begin{aligned} 0 = & \sum_{1 \leq i \leq j \leq t} (\alpha'_{jkl}{}^{(1)} - \alpha'_{jkl}{}^{(2)}) g''_j g''_k + \\ & \sum_{j \in \{1, \dots, t\}, k \in \{1, \dots, n-t\}} (\beta'_{jkl}{}^{(1)} - \beta'_{jkl}{}^{(2)}) g''_j y_k + \\ & \sum_{1 \leq i \leq j \leq n-t} (\gamma'_{ijl}{}^{(1)} - \gamma'_{ijl}{}^{(2)}) y_i y_j + \\ & \sum_{i=1}^t (\delta'_{il}{}^{(1)} - \delta'_{il}{}^{(2)}) g''_i + \\ & \sum_{j=1}^{n-t} (\nu'_{jl}{}^{(1)} - \nu'_{jl}{}^{(2)}) y_j + \\ & (\mu_l^{(1)} - \mu_l^{(2)}). \end{aligned}$$

Recall that $(g_1''(\cdot, y), \dots, g_t''(\cdot, y))$ is bijective over $\mathbb{F}_q^t$. Hence each g_j'' can be considered as an independent variable in the above equation. Since a quadratic polynomial induces the zero-mapping under evaluation if and only if it is zero itself, the claim follows if we exclude exceptional cases over the field $\mathbb{F}_2$.

To compute the coefficients in Equation (3) one simply has to evaluate $g_1'', \dots, g_m''$. Plugging the resulting values for $y_1, \dots, y_{n-t}, g_1'', \dots, g_m''$ into the equation yields linear equations in $\alpha'_{jkl}, \beta'_{jkl}, \gamma'_{ijl}, \delta'_{il}, \nu'_{jl}, \mu'_l$. Since the solution is unique we expect this system to have full rank after having generated $\mathcal{O}(n^2)$ equations this way for different inputs. Hence this step has a complexity of $\mathcal{O}((m-t)n^6)$.

Since the solution is unique it follows that the forms

$$\sum_{1 \leq i \leq j \leq n-t} \gamma'_{ijl} y_i y_j$$

are linear combinations of those quadratic parts of the OV polynomials from U , which are isometric to OV polynomials in $y_1, \dots, y_{n-t}$ as in Lemma 2.6, where $v = s$. Thereby the isometry L is defined by the representation matrix of $T \circ T^{-1'}$ restricted to the lower right $n-t$ rows and columns, which is given by C . Assuming that $\frac{q^s-1}{q^{n-t}} ((n-t-s)q^s \cdot \mathcal{P}(q, n-t-s)^{-1}) \ll 1$, which is the case for the practical instances proposed by the designers, Theorem 2.7 can be applied to these forms. Extending the resulting isometry $\tilde{C}^{-1}$ in $n-t$ to

$$C^{-1'} = \begin{pmatrix} E_t & 0 \\ 0 & \tilde{C}^{-1} \end{pmatrix} \in M(n \times n, \mathbb{F}_q),$$

E_t the $t \times t$ identity matrix, and composing the latter with G'' yields $m-t$ polynomials of the form

$$\begin{aligned} g_l''' = & \sum_{1 \leq j \leq k \leq t} \alpha''_{jkl} g_j''' g_k''' + \\ & \sum_{j \in \{1, \dots, t\}, k \in \{1, \dots, n-t\}} \beta''_{jkl} g_j''' y_k + \\ & \sum_{1 \leq i \leq j \leq s} \gamma''_{ijl} y_i y_j + \\ & \sum_{1 \leq i \leq s} y_i \ell_{il}(y_{s+1}, \dots, y_{n-t}) \\ & \sum_{i=1}^t \delta''_{il} g_i''' + \\ & \sum_{j=1}^{n-t} \nu''_{jl} y_j + \\ & \mu'_l, \end{aligned}$$

for properly chosen $\alpha''_{jkl}, \beta''_{jkl}, \gamma''_{ijl}, \delta''_{il}, \nu''_{jl}, \mu'_l$. Thereby $g_1''', \dots, g_t'''$ remains of the shape $\ell'_i(x) - \mathbf{q}_i'''(y)$ for a linear mapping ℓ'_i and quadratic mapping $\mathbf{q}_i'''$.

Obviously Step 3 is dominated by the first part, which is in $\mathcal{O}(n^6(m-t))$, or the last part, which needs $\mathcal{O}((m-t)(n-t-s)q^s \cdot \mathcal{P}(q, n-t-s)^{-1})$ quadratic form evaluations by Theorem 2.7. Since each quadratic form evaluation cost $(n-t)^2$ base field operations, this step has overall complexity of

$$\max \{ \mathcal{O}(n^6(m-t)), \mathcal{O}((m-t)(n-t)^2(n-t-s)q^s \cdot \mathcal{P}(q, n-t-s)^{-1}) \}$$

base field operations on average.

4.4 Decrypting Any Ciphertext or Forging Any Signature

Without loss of generality, we restrict to show how to attack the encryption algorithm. For an intercepted c , the mapping $S^{-1'}$ is applied resulting in $c' =$

$(c'_1, \dots, c'_m)$. For each $l = t + 1, \dots, m$, the polynomials $g_1''', \dots, g_t'''$ in the equation

$$\begin{aligned} c'_l = & \sum_{1 \leq i \leq j \leq t} \alpha''_{jkl} g_j''' g_k''' + \\ & \sum_{j \in \{1, \dots, t\}, k \in \{1, \dots, n-t\}} \beta''_{jkl} g_j''' y_k + \\ & \sum_{1 \leq i \leq j \leq s} \gamma''_{ijl} y_i y_j + \\ & \sum_{1 \leq i \leq s} y_i \ell_{il}(y_{s+1}, \dots, y_{n-t}) \\ & \sum_{i=1}^t \delta''_{il} g_i''' + \\ & \sum_{j=1}^{n-t} \nu''_{jl} y_j + \\ & \mu_l'' \end{aligned}$$

are substituted by $c'_1, \dots, c'_t$. After that, a brute-force step is conducted for all values $y_1, \dots, y_s$. For each candidate, the resulting linear system of equations in $y_{s+1}, \dots, y_{n-t}$ is solved. The corresponding solution $y_1, \dots, y_{n-t}$ is plugged into $g_1''', \dots, g_t'''$ and the remaining system of linear equations in $x_1, \dots, x_t$ is solved. This step has a complexity of

$$\mathcal{O}(q^s(m-t)(n-t-s) \cdot \min(m-t, n-t-s) + q^s t(n-t)^2 + q^s t^3)$$

base field operations. Applying $T^{-1'} C^{-1'}$ to the resulting vector in $\mathbb{F}_q^n$ and checking if the result is a correct message conforming to the redundancy r yields finally the plaintext and we are done.

The overall attack complexity for $m \leq n, t, s \leq n$ (the typical case) is summarized in the following theorem, showing that any (practical) system based on this design is completely broken.

Theorem 4.2. *Pesto with parameters $m \leq n, t, s \leq n$ and*

$$\frac{q^s - 1}{q^{n-t}} \left((n-t-s)q^s \cdot \mathcal{P}(q, n-t-s)^{-1} \right) \ll 1$$

can be broken with a precomputation of a worst-case complexity of

$$\max \left\{ \mathcal{O}(n^6(m-t)), \mathcal{O}((m-t)(n-t)^2(n-t-s)q^s \cdot \mathcal{P}(q, n-t-s)^{-1}) \right\}$$

base field operations. Deciphering an intercepted message or forging a signature has an online complexity of $\mathcal{O}(q^s(m-t)(n-t-s) \cdot \min(m-t, n-t-s) + q^s t(n-t)^2 + q^s t^3)$ base field operations.

Proof. Obviously, the complexity of Step 3 dominates the precomputation (note that we assume $m \leq n$). Moreover, deciphering or forging a signature is carried out with the substitute key almost as fast as with the original key. The additional matrix multiplications with $S^{-1'}, T^{-1'} C^{-1'}$ do not increase the overall cost, as this is dominated by solving q^s times a system in initially $n-t-s$ and then in t indeterminates. $\square$

Remark 4.2. *Note, that the online phase of our attack is about as fast as using the legitimate key for practical choices of n, t, s .*

5 Conclusion

In this paper, we break **Pesto** for practical security parameters $m \leq n, t, s \leq n$ by targeting the decomposition problem to recover an equivalent secret key.

To build possibly secure systems based on the idea introduced in [2] one has to remove the separation of variables induced by $x - \mathbf{q}(y)$ and the OV polynomials. The challenge is then to find a way to get an easy-to-invert mapping by maintaining cryptosystems with practical key sizes. Despite the attack presented here we think it is valuable to conduct further research in this direction. Having secure multivariate cryptosystems is an important cornerstone in post-quantum cryptography.

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